

SQL Server — Class 1 Homework

Querying · Sorting · Filtering · Set Operators

Instructions

Write a SQL Server query to answer each question below using the BikeStores sample database.

Database: BikeStores Tables: production.products, production.brands, production.categories, sales.customers, sales.orders, sales.order_items, sales.staffs, sales.stores, sales.stocks

Topic Legend:

Querying

Sorting

Filtering

Set Operators

Q1 List all brand names from the brands table.

Querying

Your answer:

```
select * from [production].[brands];
```

100 % No issues found

Results			Messages		
	brand_id	brand_name			
1	1	Electra			
2	2	Haro			
3	3	Heller			
4	4	Pure Cycles			
5	5	Ritchey			
6	6	Strider			
7	7	Sun Bicycles			
8	8	Surly			
9	9	Trek			

Q2

Show the product name and list price of all products, sorted from most expensive to cheapest.

Sorting

Your answer:

```
select * from [production].[products] order by [list_price] desc;
```

100 % ✓ No issues found

Results **Messages**

	product_id	product_name	brand_id	category_id	model_year	list_price
1	155	Trek Domane SLR 9 Disc - 2018	9	7	2018	11999.99
2	149	Trek Domane SLR 8 Disc - 2018	9	7	2018	7499.99
3	156	Trek Domane SL Frameset - 2018	9	7	2018	6499.99
4	157	Trek Domane SL Frameset Women's - 2018	9	7	2018	6499.99
5	169	Trek Emonda SLR 8 - 2018	9	7	2018	6499.99
6	51	Trek Silque SLR 8 Women's - 2017	9	7	2017	6499.99
7	50	Trek Silque SLR 7 Women's - 2017	9	7	2017	5999.99
8	56	Trek Domane SLR 6 Disc - 2017	9	7	2017	5499.99
9	177	Trek Domane SLR 6 Disc - 2018	9	7	2018	5499.99
10	154	Trek Domane SLR 6 Disc Women's - 2018	9	7	2018	5499.99
11	148	Trek Domane SL 8 Disc - 2018	9	7	2018	5499.99

Q3 Find all customers who live in the state of New York (NY).

Filtering

Your answer:

select * from [sales].[customers] where [state] = 'NY';

100 % ✓ No issues found

Ln: 3, Ch: 1 (55 chars, 1 li

Results **Messages**

	customer_id	first_name	last_name	phone	email	street	city	state	zip_code
1	1	Debra	Burks	NULL	debra.burks@yahoo.com	9273 Thome Ave.	Orchard Park	NY	14127
2	4	Daryl	Spence	NULL	daryl.spence@aol.com	988 Pearl Lane	Uniondale	NY	11553
3	6	Lyndsey	Bean	NULL	lyndsey.bean@hotmail.com	769 West Road	Fairport	NY	14450
4	7	Latasha	Hays	(716) 986-3359	latasha.hays@hotmail.com	7014 Manor Station Rd.	Buffalo	NY	14215
5	8	Jacqueline	Duncan	NULL	jacqueline.duncan@yahoo.com	15 Brown St.	Jackson Heights	NY	11372
6	9	Genoveva	Baldwin	NULL	genoveva.baldwin@msn.com	8550 Spruce Drive	Port Washington	NY	11050
7	10	Pamela	Newman	NULL	pamela.newman@gmail.com	476 Chestnut Ave.	Monroe	NY	10950
8	11	Deshawn	Mendoza	NULL	deshawn.mendoza@yahoo.com	8790 Cobblestone Street	Monsey	NY	10952
9	12	Robby	Sykes	(516) 583-7761	robby.sykes@hotmail.com	486 Rock Maple Street	Hempstead	NY	11550

Q4 Show only the top 5 most expensive products in the store.

Querying

Your answer:

select TOP 5 * from [production].[products] order by list_price desc ;

100 % No issues found



Results



Messages

	product_id	product_name	brand_id	category_id	model_year	list_price
1	155	Trek Domane SLR 9 Disc - 2018	9	7	2018	11999.99
2	149	Trek Domane SLR 8 Disc - 2018	9	7	2018	7499.99
3	51	Trek Silque SLR 8 Women's - 2017	9	7	2017	6499.99
4	156	Trek Domane SL Frameset - 2018	9	7	2018	6499.99
5	157	Trek Domane SL Frameset Women's - 2018	9	7	2018	6499.99

Q5

List all products with a price between \$200 and \$500, sorted by price ascending.

Filtering

Your answer:

select TOP 5 * from [production].[products] where list_price between \$200 and \$500 ;

100 % No issues found



Results



Messages

	product_id	product_name	brand_id	category_id	model_year	list_price
1	1	Trek 820 - 2016	9	6	2016	379.99
2	6	Surly Ice Cream Truck Frameset - 2016	8	6	2016	469.99
3	13	Electra Cruiser 1 (24-Inch) - 2016	1	3	2016	269.99
4	14	Electra Girl's Hawaii 1 (16-inch) - 2015/2016	1	3	2016	269.99
5	17	Pure Cycles Vine 8-Speed - 2016	4	3	2016	429.00

Q6

Find all customers whose last name starts with the letter "S".

Filtering

Your answer:

Select * from [sales].[customers] where last_name like '%s';

100 % ✓ No issues found Ln: 3, Ch: 1 (61 chars, 1 lines) TABS CRLF									
Results Messages									
	customer_id	first_name	last_name	phone	email	street	city	state	zip_code
1	1	Debra	Burks	NULL	debra.burks@yahoo.com	9273 Thome Ave.	Orchard Park	NY	14127
2	7	Latasha	Hays	(716) 986-3359	latasha.hays@hotmail.com	7014 Manor Station Rd.	Buffalo	NY	14215
3	12	Robby	Sykes	(516) 583-7761	robby.sykes@hotmail.com	486 Rock Maple Street	Hempstead	NY	11550
4	17	Caren	Stephens	NULL	caren.stephens@msn.com	914 Brook St.	Scarsdale	NY	10583
5	29	Syreeta	Hendricks	NULL	syreeta.hendricks@msn.com	193 Spruce Road	Mahopac	NY	10541
6	44	Onita	Johns	NULL	onita.johns@msn.com	32 Glen Creek Lane	Elmont	NY	11003
7	49	Caroll	Hays	NULL	caroll.hays@yahoo.com	9381 Wrangler St.	Fairport	NY	14450
8	51	Gertrud	Rhodes	NULL	gertrud.rhodes@aol.com	9961 Meadowbrook Street	Memick	NY	11566
9	63	Tenisha	Lyons	NULL	tenisha.lyons@aol.com	78 E. Rock Creek Street	Amityville	NY	11701
10	72	Petronila	Norris	NULL	petronila.norris@aol.com	72 South Marshall Lane	South El Monte	CA	91733
11	73	Melanie	Hayes	NULL	melanie.hayes@msn.com	5 N. Hall Dr.	Liverpool	NY	13090

Q7

List all products that belong to category 6 (Mountain Bikes) OR category 7 (Road Bikes).

Filtering

Your answer:

select * from [production].[products] where category_id in (6, 7);

100 % ✓ No issues found						
Results Messages						
	product_id	product_name	brand_id	category_id	model_year	list_price
1	1	Trek 820 - 2016	9	6	2016	379.99
2	2	Ritchey Timberwolf Frameset - 2016	5	6	2016	749.99
3	3	Surly Wednesday Frameset - 2016	8	6	2016	999.99
4	4	Trek Fuel EX 8 29 - 2016	9	6	2016	2899.99
5	5	Heller Shagamaw Frame - 2016	3	6	2016	1320.99
6	6	Surly Ice Cream Truck Frameset - 2016	8	6	2016	469.99
7	7	Trek Slash 8 27.5 - 2016	9	6	2016	3999.99
8	8	Trek Remedy 29 Carbon Frameset - 2016	9	6	2016	1799.99
9	27	Surly Big Dummy Frameset - 2017	8	6	2017	999.99

100 % ✓ No issues found						
Results Messages						
	product_id	product_name	brand_id	category_id	model_year	list_price
32	50	Trek Silque SLR 7 Women's - 2017	9	7	2017	5999.99
33	51	Trek Silque SLR 8 Women's - 2017	9	7	2017	6499.99
34	52	Surly Steamroller - 2017	8	7	2017	875.99
35	53	Surly Ogre Frameset - 2017	8	7	2017	749.99
36	54	Trek Domane SL Disc Frameset - 2017	9	7	2017	3199.99
37	55	Trek Domane S 6 - 2017	9	7	2017	2699.99
38	56	Trek Domane SLR 6 Disc - 2017	9	7	2017	5499.99
39	57	Trek Emonda S 5 - 2017	9	7	2017	1999.99
40	58	Trek Madone 9.2 - 2017	9	7	2017	4999.99
41	59	Trek Domane S 5 Disc - 2017	9	7	2017	2599.99

Q8

Show all orders that have NOT been shipped yet (shipped_date is missing).

Filtering

Your answer:

select * from [sales].[orders] where shipped_date is NULL;

100 %

✓ No issues found



Results



Messages

	order_id	customer_id	order_status	order_date	required_date	shipped_date	store_id	staff_id
1	70	50	3	2016-02-11	2016-02-11	NULL	3	9
2	76	9	3	2016-02-16	2016-02-16	NULL	2	6
3	158	117	3	2016-04-04	2016-04-04	NULL	3	9
4	185	24	3	2016-04-20	2016-04-20	NULL	1	2
5	239	13	3	2016-05-27	2016-05-27	NULL	3	9
6	246	17	3	2016-05-30	2016-05-30	NULL	2	6
7	264	5	3	2016-06-10	2016-06-10	NULL	1	3

Q9

Display product name, brand ID, and a computed column showing 15% off the list price, labeled "Sale Price".

Querying

Your answer:

Select product_name, brand_id, list_price * 0.85 AS "Sale Price"
from [production].[products]

100 %

✓ No issues found



Results



Messages

	product_name	brand_id	Sale Price
1	Trek 820 - 2016	9	322.9915
2	Ritchey Timberwolf Frameset - 2016	5	637.4915
3	Surly Wednesday Frameset - 2016	8	849.9915
4	Trek Fuel EX 8 29 - 2016	9	2464.9915
5	Heller Shagamaw Frame - 2016	3	1122.8415
6	Surly Ice Cream Truck Frameset - 2016	8	399.4915
7	Trek Slash 8 27.5 - 2016	9	3399.9915
8	Trek Remedy 29 Carbon Frameset - 2016	9	1529.9915
9	Trek Conduit+ - 2016	9	2549.9915
10	Surly Straddler - 2016	8	1316.6500

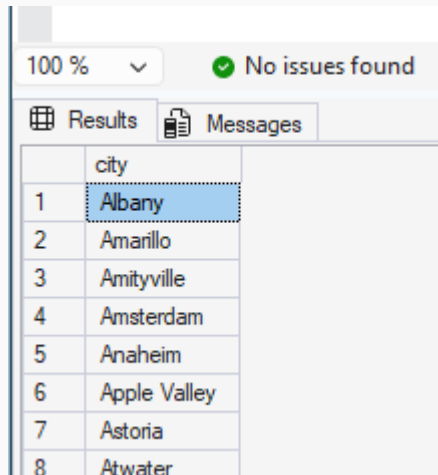
Q10

Get a unique list of all cities where BikeStores customers live, sorted alphabetically.

Querying

Your answer:

select DISTINCT city from [sales].[customers] ORDER BY city ASC;



100 % No issues found

Results Messages

	city
1	Albany
2	Amarillo
3	Amityville
4	Amsterdam
5	Anaheim
6	Apple Valley
7	Astoria
8	Atwater

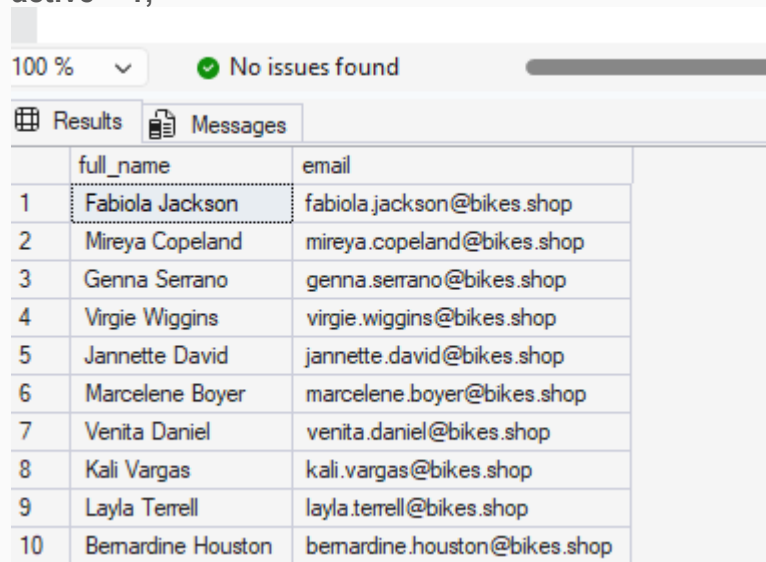
Q11

List all staff members who are currently active (active = 1), showing their full name and email.

Filtering

Your answer:

select first_name || ' ' || last_name AS full_name , email from [sales].[staffs] WHERE active = 1;



100 % No issues found

Results Messages

	full_name	email
1	Fabiola Jackson	fabiola.jackson@bikes.shop
2	Mireya Copeland	mireya.copeland@bikes.shop
3	Genna Serrano	genna.serrano@bikes.shop
4	Virgie Wiggins	virgie.wiggins@bikes.shop
5	Jannette David	jannette.david@bikes.shop
6	Marcelene Boyer	marcelene.boyer@bikes.shop
7	Venita Daniel	venita.daniel@bikes.shop
8	Kali Vargas	kali.vargas@bikes.shop
9	Layla Terrell	layla.terrell@bikes.shop
10	Bernardine Houston	bernardine.houston@bikes.shop

Q12

Using UNION, get a combined list of all cities from both customers and stores (no duplicates), sorted A–Z.

Set Operators

Your answer:

Select city from sales.customers union select city from sales.stores;

100 %	✓ No issues found
Results	Messages
	city
1	Albany
2	Amarillo
3	Amityville
4	Amsterdam
5	Anaheim
6	Apple Valley
7	Astoria
8	Atwater
9	Auburn

Q13

Using EXCEPT, find product IDs that exist in order_items but are NOT in the products table.

Set Operators

Your answer:

**Select product_id From sales.order_items
EXCEPT
Select product_id from production.products;**

100 %	✓ No issues found
Results	Messages
	product_id

Q14

List all products from the year 2016 with a price greater than \$1,500, sorted by price descending.

Filtering

Your answer:

**select product_name from production.products where list_price > \$1500 and
model_year = 2016 ;**

100 %  No issues found

 Results  Messages

	product_name
1	Trek Fuel EX 8 29 - 2016
2	Trek Slash 8 27.5 - 2016
3	Trek Remedy 29 Carbon Frameset - 2016
4	Trek Conduit+ - 2016
5	Surly Straggler - 2016
6	Surly Straggler 650b - 2016

Q15

BONUS: Using UNION ALL, show Trek (brand_id = 9) products from 2016 and Surly (brand_id = 8) products from 2017. Include product name, brand, year, and price. Order by year then price descending.

Bonus

Your answer:

```
SELECT
  product_name,
  brand_id AS brand,
  model_year AS year,
  list_price AS price
FROM production.products
WHERE brand_id = 9 AND model_year = 2016
```

UNION ALL

```
SELECT
  product_name,
  brand_id AS brand,
  model_year AS year,
  list_price AS price
FROM production.products
WHERE brand_id = 8 AND model_year = 2017
```

```
ORDER BY year, price DESC;
```

100 %		No issues found		
Results		Messages		
	product_name	brand	year	price
1	Trek Slash 8 27.5 - 2016	9	2016	3999.99
2	Trek Conduit+ - 2016	9	2016	2999.99
3	Trek Fuel EX 8 29 - 2016	9	2016	2899.99
4	Trek Remedy 29 Carbon Frameset - 2016	9	2016	1799.99
5	Trek 820 - 2016	9	2016	379.99
6	Surly Karate Monkey 27.5+ Frameset - 2017	8	2017	2499.99
7	Surly Wednesday - 2017	8	2017	1632.99
8	Surly Ice Cream Truck Frameset - 2017	8	2017	999.99
9	Surly Big Dummy Frameset - 2017	8	2017	999.99
10	Surly Steamroller - 2017	8	2017	875.99
11	Surly Troll Frameset - 2017	8	2017	832.99